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Apparatuses, systems, and methods for seed double elimination in agricultural operations

Abstract

An example seed double eliminator includes at least one tine that is disposed adjacent to an inner surface of a seed disc. The at least one tine is pivotably movable about an axis of rotation to engage seed doubles located at one or more locations along the inner surface of the seed disc and eliminate excess seeds such that a single seed is retained at the location. The at least one tine is pivotably movable in response to operation of a position selector to cause an angular orientation of the tine to become altered.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to eliminating excess seed during an agricultural planting operation.

BACKGROUND OF THE DISCLOSURE

(2) Agricultural planting operations involve depositing seed into the ground. Generally, planting operations involve the use of seed meters that utilize a pressure differential (such as a vacuum or positive pressure) to adhere seed to an aperture formed in a seed meter. The seed meter operates to select individual seeds that are subsequently carried from the seed meter to the ground and deposited into the soil, such as in a furrow formed in the soil.

SUMMARY OF THE DISCLOSURE

(3) A first aspect of the present disclosure is directed to a seed double eliminator for a seeding machine having a seed disc. The seed double eliminator may include a housing that includes a cavity and a shaft formed in the cavity, the shaft defining a pivot axis. The seed double eliminator may also include a tine mounted on the shaft and pivotable about the pivot axis and a selector operably engaged with the tine. The selector may be movable to alter an angular orientation of the tine about the pivot axis.

(4) A second aspect of the present disclosure is directed to an agricultural planter. The agricultural

planter may include a seeding machine configured to singulate seeds prior to depositing the seeds into the ground. The seeding machine may include a rotatable seed disc and a seed double eliminator disposed adjacent to an inner surface of the seed disc. The seed double eliminator may include a housing disposed along a perimeter of the seed meter. The housing may include a cavity and a shaft extending into the cavity and defining a pivot axis. The seed double eliminator may also include a tine mounted on the shaft and pivotable about the pivot axis and a selector operably engaged with the tine. The selector may be movable to alter an angular orientation of the tine about the pivot axis.

(5) Another aspect of the present disclosure is directed to a method for altering a rotational orientation of a seed double eliminator. The method may include operating a selector of a seed double eliminator. The seed double eliminator may include the selector, a frame operably engaged with the selector and a tine pivotably movable about a first axis of rotation. The frame may be operably engaged with the tine to cause the pivotable movement of the tine. The method may also include moving the frame in response to operation of the selector and imparting a moment to the tine by the frame to pivot the tine about the first axis of rotation.

(6) The various aspects of the present disclosure may include one or more of the following features. The selector may be movable to cause movement of the tine about the pivot axis in discrete amounts. The selector may be movable to cause movement of the tine over a predetermined range. The range of movement may be approximately 30° of rotation. A frame may extend between the tine and the selector, and the frame may be movable in response to movement of the selector to cause movement of the tine about the pivot axis. The tine may include an elongated portion, a body portion, and a protrusion extending from the body portion. The body portion may define an aperture that receives the shaft, and the frame may include an opening. The protrusion may be received into the opening of the frame. The tine may be movable about the pivot axis in response to a moment imparted to the protrusion by the frame. The opening may define an edge of the frame, and the edge of the frame may impart the moment to the protrusion to alter the angular orientation of the tine about the pivot axis. The selector may be rotatable, and rotation of the selector may cause movement of the frame along a path. The selector may include a first gear; the frame may include a second gear intermeshed with the first gear; and rotation of the selector may cause the first gear to rotate the second gear which, in turn, may cause movement of the frame along the path. The selector may include a powered actuator, and actuation of the actuator may cause movement of the frame along a path. The frame may be disposed in a channel, and movement of the selector may cause the frame to move along a curved path within the channel.

(7) Additionally, the various aspects may include one or more of the following features. The seed double eliminator may include a frame extending between the tine and the selector. The frame may be movable in response to movement of the selector to cause movement of the tine about the pivot axis. The frame may be disposed in a channel, and movement of the selector may cause the frame to move along a curved path within the channel. The tine may include an elongated portion and a body portion that includes a protrusion and defines an aperture that receives the shaft. The frame may include an opening, and the protrusion may be received into the opening of the frame. The tine may be movable about the pivot axis in response to a moment imparted to the protrusion by the frame. The selector may be rotatable, and rotation of the selector may cause movement of the frame along a path. The selector may include a powered actuator, and actuation of the actuator may cause movement of the frame along a path. Moving the frame in response to operation of the selector may include moving the frame along a path in response to operation of the selector. The selector may include a powered actuator operably engaged with the frame. Operating the selector may include actuating the actuator, and moving the frame in response to operation of the selector may include displacing the frame in response to actuation of the actuator. The frame may include an aperture, and a protrusion of the tine may be received into the aperture. Imparting a moment to the tine by the frame to pivot the tine about the second axis of rotation may include imparting a force to the

protrusion by surface of the frame defining the aperture.

(8) Further, the various aspects of the present disclosure may include one or more of the following features. Operating a selector may include rotating the selector about a second axis of rotation.

(9) Other features and aspects will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description of the drawings refers to the accompanying figures in which:

(2) FIG. 1 is a perspective view of an agricultural planter, according to some implementations of the present disclosure.

(3) FIG. 2 is a perspective view of a planting unit frame, seed meter, and seed delivery system, according to some implementations of the present disclosure.

(4) FIG. 3 is a detailed view of the seed meter of FIG. 2, along with the associated delivery system drives.

(5) FIG. 4 is a perspective view of the seed meter of FIG. 2 with the cover open illustrating an example seed disc of the seed meter.

(6) FIG. 5 is an exploded perspective view of the seed disc included in the seed meter of FIG. 2.

(7) FIG. 6 is cross-sectional view of the seed disc of FIG. 5.

(8) FIG. 7 is a partial cross-sectional view of the example seed meter and seed delivery of FIG. 2 showing details of the seed meter and seed delivery system.

(9) FIG. 8 is a partial cross-sectional view of an ejector disposed adjacent to an exterior surface of seed meter with a delivery system brush belt omitted.

(10) FIG. 9 is a detail view of a wheel of the ejector of FIG. 8 illustrating engagement between a protrusion of the wheel of the ejector and an adjacent seed disc.

(11) FIG. 10 is a perspective view of the seed meter of FIG. 2 showing internal components thereof, including an example seed double eliminator.

(12) FIG. 11 is a detail view of a seed disc and an example seed double eliminator disposed adjacent to the seed disc, according to some implementations of the present disclosure.

(13) FIGS. 12 through 17 are cross-sectional views of the seed double eliminator of FIG. 11 illustrating a tine of the seed double eliminator adjusted to different angular orientations.

(14) FIG. 18 is a perspective view of the seed double eliminator of FIG. 11 showing the engagement between a plurality of tines and a frame whose position is adjustable in response to operation of a position selector.

(15) FIGS. 19 and 20 are perspective views showing aspects of the seed double eliminator of FIG. 11.

(16) FIG. 21 is a detail view of a selector of the seed double eliminator of FIG. 11.

(17) FIG. 22 is a perspective view of another example seed-double eliminator of FIG. 11 in which movement of a frame is altered by an actuator.

(18) FIGS. 23 and 24 are oblique views of another example seed double eliminator, according to some implementations of the present disclosure.

(19) FIG. 25 is a cross-sectional view of the seed double eliminator of FIGS. 23 and 24.

(20) FIG. 26 is a side view of another example seed double eliminator, according to some implementations of the present disclosure.

(21) FIG. 27 is an oblique view of the seed double eliminator of FIG. 26.

(22) FIG. 28 is a cross-sectional view of another example seed double eliminator, according to some implementations of the present disclosure.

(23) FIG. 29 is a perspective view of the seed double eliminator of FIG. 28.

- (24) FIGS. 30 through 34 are perspective view of the seed double eliminator of FIG. 28, illustrating how a wheel moves in response to rotation of a seed disc and how a protrusion of the wheel obstructs an aperture formed in the seed disc.
- (25) FIG. 35 is a cross-sectional view of another example seed double eliminator, according to some implementations of the present disclosure.
- (26) FIGS. 36 through 38 are side views of a wheel of the seed double eliminator of FIG. 35 illustrating how a protrusion of the wheel obstructs an aperture formed in a seed disc as the wheel rotates about an axis.
- (27) FIG. 39 is a partial cross-sectional view of another seed double eliminator, according to some implementations of the present disclosure.
- (28) FIG. 40 is a detail view of a portion of the seed double eliminator of FIG. 39 illustrating how a lever obstructs a portion of an aperture of a seed disc.
- (29) FIG. 41 is a detail view of another example seed double eliminator, according to some implementations of the present disclosure.
- (30) FIG. 42 is a detail view of the seed double eliminator of FIG. 41 illustrating how a spring strip obstructs a portion of an aperture formed in a seed disc.
- (31) FIG. 43 is a top view of another example seed double eliminator, according to some implementations of the present disclosure.
- (32) FIG. 44 is a side view of the seed double eliminator of FIG. 43.
- (33) FIG. 45 is a detail view of a portion of the seed double eliminator shown in FIG. 25.
- (34) FIG. 46 is an oblique view of a portion of the seed meter of FIG. 2.
- (35) FIG. 47 is a side view of an example seed double eliminator having a wheel with protrusions having flattened ends.
- (36) FIG. 48 is a side view of another example seed double eliminator having a wheel with protrusions with a conical shape.

DETAILED DESCRIPTION

(37) For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the implementations illustrated in the drawings, and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is intended. Any alterations and further modifications to the described devices, systems, or methods and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one implementation may be combined with the features, components, and/or steps described with respect to other implementations of the present disclosure.

(38) Words of orientation, such as “up,” “down,” “top,” “bottom,” “above,” “below,” “leading,” “trailing,” “front,” “back,” “forward,” and “rearward” that are used in the context of the illustrated examples are used as would be understood by one skilled in the art and are not intended to be limiting to the disclosure. For example, for a particular type of vehicle or implement in a conventional configuration and orientation, one skilled in the art would understand these terms as the terms apply to the particular vehicle or implement. Additionally, although terms such as “upper” and “lower” may be used to describe features as depicted in the various figures, these are not intended to limit the disclosure to the particular orientation depicted. For example, in some cases, at a particular orientation, a feature described herein as “upper” may be located vertically relative to, but to the right (or left) of a feature described herein as “lower.”

(39) For example, as used herein, with respect to a work vehicle, unless otherwise defined or limited, the term “forward” (and the like) corresponds to a forward direction of travel of the work vehicle over the ground during normal operation of the work vehicle. Likewise, the term “rearward” (and the like) corresponds to a direction opposite the forward direction of travel of the work vehicle.

(40) Also as used herein, with respect to an agricultural implement or components thereof, unless otherwise defined or limited, the term “leading” (and the like) indicates a direction of travel of the agricultural implement when viewed in a conventional orientation on flat ground during normal operation (e.g., the forward direction of travel of a work vehicle transporting an implement). Similarly, the term “trailing” (and the like) indicates a direction that is opposite the leading direction. A conventional orientation represents a work vehicle being oriented such that normal operation of the work vehicle can be performed. For example, a conventional orientation may involve having the tracks or wheels of the vehicle or field engaging components of an implement contacting the ground in a manner that allows the work vehicle or implement to function as intended.

(41) A disc-shaped or bowl-shaped seed meter within a planting unit moves seed along a generally circular seed path from a seed pool to an elevated release position. Seed meters having other shapes and seed paths having other shapes are also contemplated. Seed doubles may sometimes be formed on such a seed meter. A seed double is any grouping of multiple seeds, including groupings of three or more seeds, present at a location on the seed meter intended to secure a single seed. These locations generally correspond to apertures formed in a seed disc of the seed meter that are intended to carry one seed at a time during a planting operation. In instances in which a seed double is present at one or more of the locations, the seed doubles are carried along the circular path, potentially resulting in sub-optimal seed delivery (e.g., poor or failed transport for final planting by a seed delivery system), ineffective seed singulation, and seed waste. A seed-double eliminator is used to assist in a more efficient and successful seed singulation by removing any excess seeds from various seed doubles on the seed meter.

(42) Seeds within a seed double may extend farther away from the seed meter than a single seed being carried by the seed meter, providing an avenue for removal of excess seeds while retaining a single seed at the aperture. In addition to other benefits, such flexible members may take advantage of this phenomenon in order to eliminate seed doubles more selectively.

(43) FIG. 1 is a perspective view of an example agricultural seeding machine **10**. In the illustrated example the seeding machine **10** is a row crop planter. It will be understood that various other configurations may also be possible and that the various seed double eliminators disclosed herein may be used in a variety of agricultural machinery or other settings. The seeding machine **10** includes a central frame **12** on which a plurality of individual planting units **14** are mounted. The seeding machine **10** is oriented with a fore-aft direction shown by arrow **15** and a transverse direction shown by arrow **17**. Each planting unit **14** is coupled to the central frame **12**, such as by a parallel linkage (e.g., linkage **16**), so that individual planting units **14** can move up and down relative to frame **12**. Large storage tanks **13** hold seed that is delivered pneumatically to a mini hopper on each planting unit.

(44) FIG. 2 shows an example planting unit **14**. Each planting unit **14** can be mounted, in various ways, to the central frame **12**. The planting unit **14** is provided merely as an example. Consequently, the scope of the present disclosure encompasses the use or inclusion of the various seed double eliminators described herein with various other seed-handling devices. A frame **18** of the planting unit **14** includes a pair of upstanding arms **20** at the forward end thereof. The arms are coupled to the rearward ends of parallel linkage **16**. A shaft **22** is coupled to the frame **18**, and furrow opening disks **28**, shown in FIG. 1, are attached to the shaft **22** and operate to form an open furrow in the soil beneath the seeding machine. The planting unit **14** deposits seed into the furrow created by the opening disks **28**. Closing and packing wheels **32**, also shown in FIG. 1, are mounted to the frame **18** and operate to close the furrow over the deposited seed. In some instances, the packing wheels **32** also firm the soil in the closed furrow. In some instances, closing wheels are included that operate to close a furrow after delivery of the seeds. A seed meter **24** and seed delivery system **26** are also attached to the frame **18** of the planting unit **14**.

(45) Referring to FIGS. 3 and 4, the seed meter **24** includes a housing **30** and a cover **34**. The

housing **30** and the cover **34** are coupled to one another by a complementary hinge features included on housing **30** and cover **34**, respectively, that combine to form a hinge **33**. The seed meter **24** also includes an electric motor **40** and a drive spindle **36** carried by the housing **30**. The drive spindle **36** is coupled to an output shaft **38** of the electric motor **40**. The electric motor **40** is used to operate the seed meter **24**. More particularly, the electric motor **40** rotates a seed disc of the seed meter **24** about an axis of rotation **37**.

(46) The planting unit **14** also includes an electric motor **42** that drives the delivery system **26**. In the illustrated example, an output shaft of the electric motor **42** is connected to the delivery system **26** via a right-angle drive **44**. However, in other implementations, other arrangements are used to provide motive power from the electric motor **42** to the delivery system **26**. Further, while electric motors have been shown to drive both the seed meter **24** and the seed delivery system **26**, it will be appreciated by those skilled in the art that other types of motive devices, such as hydraulic motors or pneumatic motors, can be used. Further, other types of mechanical drive systems may also be used.

(47) Referring also to FIGS. **5** and **6**, a seed disc **46** of seed meter **24** is shown in greater detail. The seed disc **46** operates to selectively retain individual seeds from a seed pool located within the seed meter **24** at selected locations along the seed disc **46** and transport the individual seeds to the delivery system **26**. The delivery system **26** transports the seeds to the ground for planting. In the illustrated example, the seed disc **46** is bowl-shaped. However, the scope of the present disclosure is not limited to bowl-shaped seed discs. Rather, the present disclosure encompasses other types of seed discs that operate to transport individual seeds from one location to another. For example, other types of seed discs within the scope of the present disclosure include seed discs that are flat or have a generally flat shape (commonly referred to as seed plates). Thus, although the various seed double eliminators described herein are described in the context of a bowl-shaped seed disc, seed discs having other configurations are included within the scope of the present disclosure.

(48) As explained, the example seed disc **46** has a concaved bowl-shape defining an interior space **47**. The seed disc **46** includes a base portion **48** from which a conical side wall **50** extends. The side wall **50** terminates in a peripheral edge (referred to hereinafter as outer edge **52**). The side wall **50** includes a rim portion **54** that is adjacent to the outer edge **52**. The rim portion **54** is indicated generally by a bracket in FIGS. **5** and **6**. In the illustrated example, the rim portion **54** extends along a portion of the side wall **50** from the outer edge **52**. An annular array of apertures **56** is located within the rim portion **54**. The apertures **56** extend between an inner surface **55** and an outer surface **57** of the side wall **50**.

(49) The seed disc **46** is mounted in the housing **30** for rotation about the axis of rotation **37** in the direction of arrow **58**, as shown in FIG. **5**. In operation, as the seed disc **46** rotates, individual seeds from a seed pool **60** present in a lower portion of the interior space **47** adhere to the apertures **56** along the inner surface **55** of side wall **50**. As the seed disc **46** rotates, the individual seeds are sequentially carried upwards to a release position **62** located at an upper portion of seed disc **46**. A series of raised features or projections (referred to hereinafter as paddles **64**) extend from the inner surface **55**. In some implementations, a paddle **64** is located adjacent to each aperture **56**. In the context of the direction of rotation **58**, a paddle **64** is located behind each aperture **56**. Each paddle **64**, accordingly, forms a confronting surface **66** behind the associated aperture **56** in the direction of rotation **58** to push a seed adhered to the aperture **56** into delivery system **26** as described below. In some instances, the seed disc **46**, as installed in housing **30**, is oriented at an angle from vertical, as illustrated, for example, in FIG. **7**.

(50) In some implementations, the seed disc **46** includes a raised feature **68** that is positioned between a paddles **64** and a second type of paddle **104**. The paddle **104** includes a V-shaped notch **105**. The paddles **104** operate to prevent seed from being collected or becoming lodged at the locations of the paddles **104**. As discussed in more detail below, the paddles **64** assist in guiding seeds into a seed transport apparatus of the seed delivery system **26**. In operation, the raised feature

68 sometimes serves to orient individual seeds in a seed double for improved removal by a seed double eliminator.

(51) In some implementations, the seed disc **46** is a one piece or unitary component. In other implementations, the seed meter **46** is constructed of multiple pieces. Further, in some implementations, a seed disc within the scope of the present disclosure may differ in various ways from the example seed disc **46** depicted in the various figures herein. It will be understood, accordingly, that the various seed double eliminators described herein are usable with a seed disc, such as seed disc **46**, or with various other types of seed transport devices.

(52) As noted above, the seed pool **60** is formed in the interior space **47** at the bottom of seed disc **46**. A vacuum is applied at the outer surface **57** of side wall **50**, causing individual seeds to be adhered to the various apertures **56** as the apertures **56** travel through the seed pool **60**. As the seed disc **46** rotates in the direction of arrow **58**, seed adheres at the apertures **56**. As the seed disc **46** continues to rotate, the adhered seed is moved upwards to the release position **62** at the upper portion of seed disc **46**. In some implementations, the release position **62** is located slightly past the top or 12 o'clock position along a circular path of travel of the seed defined by rotation of the seed disc **46** such that the seed is moving somewhat downward at the release position **62**.

(53) As shown in FIG. 7, the seed delivery system **26** is positioned adjacent to the inner wall and beneath the upper portion of seed disc **46** at the release position **62** to take seed **63** from the seed disc **46**. It will be understood that other orientations of the seed **46** (or other components) are possible and within the scope of the present disclosure.

(54) With continued reference to FIG. 7, the delivery system **26** includes a housing **70** partially enclosing a continuous seed transport apparatus **72**. In the illustrated example the continuous seed transport apparatus **72** is a brush belt that includes a plurality of bristles **73**. In other implementations, the seed transport apparatus **72** is an endless flighted belt or another device operable to transfer seed **63** from the seed disc **46** to the ground continuously.

(55) The seed transport apparatus **72** detaches the seed **63** from the seed disc **46** and transports the seed **63** to the ground. In the context of FIG. 7, the seed transport apparatus **72** generally travels in a clockwise direction within the housing **70**. An upper opening **74** is formed in the housing **70** in order to allow the seed **63** to enter the housing **70** from the seed disc **46**. The bristles **73** extend through the opening **74** in order to receive the seed **63** from seed disc **46**. As also noted above, it will be understood that other configurations are within the scope of the present disclosure. For example, in some instances, the delivery system **26** may be oriented horizontally or at an angle otherwise deviating from vertical, and the opening **75** may be generally viewed as an inlet opening to delivery system **26**.

(56) In some instances, an ejector **76** rides on the outer surface **57** of seed disc **46**, with projections from a star wheel on the ejector **76** extending, sequentially, into the apertures **56** in order to force seed **63** away from or out of the apertures **56** so that the seed can be received by the seed transport apparatus **72**. Thus, the ejector **76** is located adjacent to the outer surface **57** of the seed disc **46** at the location of the release point **62**. In some instances, the ejector **76** is biased against the outer surface **57** of the seed disc **46**, such as with a spring, and, in response to the rotation of the seed disc **46**, the star wheel of the ejector **76** “walks” along seed disc **46** such that successive projections of the star wheel sequentially eject or fully separate the seeds **63** from successive apertures **56** at release position **62**. These ejected seeds **63** are captured by the seed transport apparatus **72** (such as within the plurality of bristles **73** of the seed transport apparatus **72**) and are carried to a seed ejection point **78**.

(57) FIG. 8 shows a detailed portion of the seed meter **24** showing interaction between the seed disc **46** and the ejector **76**. The seed transport apparatus **72** is omitted for clarity. The ejector **76** includes a base **80**, a pivot arm **82** pivotably mounted to the base **80**, and biasing component **84** (e.g., a spring) that biases the pivot arm **82** towards the outer surface **57** of the seed disc **46**. The ejector **76** also include a star wheel **86** rotatably coupled to the pivot arm **82**. The star wheel **86**

include radially extending protrusions **88** that extend into the apertures **56** formed in the seed disc **46** as the star wheel **86** rotates about an axis **90** as the seed disc **46** rotates. As shown in FIG. **9**, the protrusion **88** of the star wheel **86** extends into and through the aperture **56** so as to release the seed **63** from the aperture **56**. The protrusion **88** extends beyond the inner surface **55** by a distance **92** that ensures separation of the seed **63** from the seed disc **46**. In some implementations, the distance **92** is within a range of approximately 2.0 millimeters (mm) (0.08 inches (in.)) to 5.0 mm (0.20 in.). Further, in some implementations, a size of the protrusions **88** (e.g., a width of the protrusions **88**) is selected so that insertion of the protrusions **88** into the apertures **56** occludes the apertures **56** to an extent so as to cease or reduce the applied vacuum to cause the seed **63** to separate from the seed meter **46**. For example, in some instances, the protrusions **88** occlude an open area defined by the apertures **56** by between approximately 75% to 100%. The occlusion of the apertures **56** reduces the applied vacuum, which reduces a holding force between the seed and the seed disc **46**. In some instances, the applied vacuum force is reduced to approximately zero. In addition to a reduction in applied vacuum, an amount of time of that the vacuum is reduced also assists in releasing seeds from the apertures **56** at the release position **62**.

(58) As noted above, seed disc **46** is intended to carry seeds individually and sequentially between the seed pool **60** and the release position **62** (e.g., to carry a single seed **63** between each pair of paddles **64**, secured by vacuum applied through the associated aperture **56**). In various instances, however, multiple seeds **63** from the seed pool **60** become lodged between a pair of paddles **64** or otherwise adhered to a single aperture **56**. As noted above, this condition is referred to as a “seed double” (although, in various instances, more than two seeds may be included). The presence of such seed doubles detrimentally affects the efficiency and efficacy of a planting operation, such as by planting one or more undesired seeds at a particular location in the ground. Seed doubles also result in waste of seed.

(59) The following portion of the description describes different seed double eliminators that operate to release one or more excess seeds from an aperture to ensure a single seed is present at the apertures formed in a seed disc by the time the seed reaches a delivery system of a seed meter.

(60) FIG. **10** is a perspective view of the housing **30** of the seed meter **24** with the seed disc **46** omitted to show a first type of seed double eliminator **100**. The seed-double eliminators **100** are arranged along a side **102** of the housing **30** and adjacent to the inner surface **55** of the seed disc **46**.

(61) FIG. **11** is detail view of a portion of the seed disc **46** and a portion of the housing **30** of the seed meter **24**. As explained earlier, the seed disc **46** includes a plurality of apertures **56** formed along the rim portion **54**. Each aperture **56** defines an opening area **101** (indicated by the cross-hatching), defining an area of an opening of the apertures **56**, such as where the aperture **56** intersects the inner surface **55**. A first paddle **64**, a second paddle **104**, and a raised feature **68** are disposed between adjacent apertures **56**. The first paddles **64** and second paddles **104** and the raised features **68** extend inwardly from the inner surface **55** of the seed disc **46**. Arranged adjacent to the outer edge **52** is a plurality of adjustable tines **106**. Each tine **106** extends towards the rim portion **54** along the inner surface **55** of the seed disc **46**. The tines **106** operate to remove excess seeds from seed doubles so that a single seed is retained at an aperture **56**.

(62) As the seed disc **46** rotates in the direction of rotation **58**, the tines **106** engage the seed or seeds that are retained at the location of each aperture **56**. The tines **106** disrupt a position of or otherwise unsettle the seed or seeds, and, if a seed double is located at a particular aperture **56**, the disruption caused by the tines **106** is sufficient to cause any excess seed (e.g., any seed in excess of one seed) to be dislodged. The released seed separates from the location of the aperture **56** and returns to the seed pool **60**, described earlier. In this way, seeds are conserved, and a single seed is retained at each aperture **56** for transfer to the delivery system **26** and subsequent transportation to the ground.

(63) The tines **106** are retained in cavities **108** formed in the housing **30** or in a body received into the housing **30**. The tines **106** are pivotable within the cavities **108** to alter an angular orientation of

the tines **106**, which affect an amount by which the tines **106** engage the seed located at the apertures **56**. In the illustrated example, four tines **106** are shown. However, in other instances, additional or fewer tines **106** are provided.

(64) FIGS. **12** through **17** show a tine **106** adjusted to different angular orientations by pivoting the tines **106** about a pivot axis **110**. The tines **106** are movable over a range of positions. Particularly, FIGS. **12** through **18** show the tine **106** at a 0° position (FIG. **12**) and at a 25° position (FIG. **18**). In the illustrated example, the tines **106** are movable over an angular range of approximately 25° . In some implementations, the angular range may exceed 25° (e.g., adjustable over a 30° range or greater); in other implementations, the angular range may be less than 25° . In some implementations, the tines **106** are moveable individually. In some implementations, the tines **106** are movable together synchronously. Thus, in some implementations, the tines **106** are movable together such that each tine **106** has the same angular orientation as the other tines **106**. In some instances, the tines are movable in 1° , 5° , or **10** increments. In other implementations, the tines **106** are movable in any selected increment. In other implementations, the tines **106** are moved in unison in one or more groups. In some implementations, the tines **106** for a single group move together in response to alterations to orientation. In other implementations, the tines **106** are arranged in two or more groups, and the tines **114** of each group are adjustable independently in unison.

(65) FIG. **18** is a perspective view of the seed double eliminator **100**. In the illustrated example, the double seed eliminator **100** includes four tines **106** having an angular orientation that is adjustable. As explained earlier, in other implementations, additional or fewer tines **106** may be included. As shown, each tine **106** includes an elongated portion **114** and a body portion **116** defining an aperture **118** and a protrusion **120** extending from the body portion **116**. In some implementations, the elongated portion **114**, body portion **116**, and the protrusion **120** form a unitary component. In other implementations, the elongated portion **114** is a separate component that is attached to the body portion **116**. In some implementations, the protrusion **120** is an integral part of the body portion **116**. In other implementations, the protrusion **120** is attached to the body portion **116**.

(66) The seed double eliminator **100** also includes a frame **122** and a selector **124** mounted to a housing **126** and coupled to the frame **312**. In the illustrated example, the frame **122** has a curved shape and resides in a slot **152** (shown in FIGS. **19** and **20**) formed in the housing **30** that is arc-shaped. In other implementations, the frame **122** and the slot **152** can have other shapes. In this example, the selector **124** is in the form of a dial that is rotated about an axis **128** to alter a position of the frame **122** relative to housing **30** in the direction of arrows **130** and **132**. The frame **122** includes openings **134** into which the protrusions **120** are received.

(67) The tines **106** are retained in the cavities **108** formed in the housing **30**. The tines **106** are mounted on shafts **136** provided in the cavities **108**. The shafts **136** are received into the apertures **118**. The tines **106** are pivotable on the shafts **136** in response to movement of the frame **122**. In operation, as the selector **124** is rotated, for example, in a first rotational direction corresponding to arrow **130**, the frame **122** moves in the direction of arrow **138**, for example. Movement of the frame **122** in the direction of arrow **138**, in response, imparts a moment to the tines **106** via interaction between the frame **122** and the protrusions **120**, causing the tines **106** to pivot on the respective shafts **136** about the pivot axis **110** in a third rotational direction corresponding to arrow **140**. As a result, a rotational orientation of the tines **106** is altered. Similarly, rotation of the selector **124** in a second rotational direction corresponding to arrow **132**, opposite the first rotational direction, causes the frame **122** to move in the direction of arrow **142**, causing the tines **106** to pivot in a fourth rotational direction corresponding to arrow **144**. Altering a position of the tines **106** in this way alters an amount by which the tines **106** interact with the seed or seeds at the apertures **56** and, in some instances, an amount by which the tines **106** extend across the apertures **56**. A degree to which the tines **106** engage the seeds and, in some instances, obstruct the apertures **56** can cause excess seeds (e.g., any number of seeds exceeding a single seed at an aperture) to become dislodged, thereby singulating seeds at the apertures **56**.

(68) FIGS. **19** and **20** are perspective views of the double seed eliminator **100** with the selector **124** and housing **126** omitted. The frame **122** includes an arm **146** having a geared surface **148** (e.g., gear teeth) that interacts with a mating gear **150** included on the rotatable selector **124**, shown in FIG. **21**. The gear **150** intermeshes with the geared surface **148** to alter a position of the frame **122** and, thus, a rotational orientation of the tines **106**.

(69) FIG. **21** is a detail view of an interface between the gear **150** of the selector **124** and the geared surface **148** of the arm **146** of the frame **122**. Rotation of the selector **124** causes translational movement of the frame **122** to alter a rotational orientation of the tine assemblies **304**, as explained earlier. As shown, the frame **122** is received within a slot **152** formed in the housing **30**. The frame **122** is slideable within the slot **152** in response to actuation of the selector **124**. As explained earlier, in some implementations, a path traveled by the frame **122** within the slot **152** is arc-shaped. That is, actuation of the selector **124** (e.g., rotation of the selector **124** in the context of implementation shown in FIGS. **19** through **21**) causes the frame **122** to slide along the arc within the arc-shaped slot **152**. In other implementations, the frame **122** slides along a linear path in response to operation of the selector **124**. Thus, in some implementations, the slot **152** has a linear shape.

(70) FIG. **22** is a perspective view of another example seed double eliminator **154** in which movement of the frame **122** is altered using an actuator **156**. The actuator **156** is operable, such as in response to input received from a user, to displace the frame **122** within the slot **152**. In some implementations, the actuator **156** is an electric actuator. However, the actuator includes other types of actuators, such as hydraulic and pneumatic actuators. In some implementations, actuation of the actuator **156** is directed via an input device, e.g., a button, dial, keyboard, slide, etc., that is available, for example, in a cab of a vehicle, on an implement carrying the **24**, or from a remote location.

(71) In the illustrated example, the actuator **156** includes a body **162** and a moveable arm **164** that is extendable and retractable from the body **162**. The arm **164** is coupled to the arm **146** of the frame **122**, such as via a pinned connection. Extension of the arm **164** in the direction of arrow **166** cause movement of the frame **122** within the slot **152** in the general direction of arrow **138**. As explained above, the frame **122** move along an arc-shaped path within the slot **152** due to the arc shape defined by the slot **152**. In other implementations, the frame **122** may move along a straight or linear path in response to actuation of the actuator **156**. In response to movement of the frame **122**, an angular orientation of the tines **106** is altered in a first rotational direction by pivoting of the tines **106** on the shafts **136**. Retraction of the arm **164** in the direction of arrow **142** causes the frame **122** to move in the general direction of arrow **142**, which, as explained may be a curved path or linear path as may be defined by the slot **152**. In response to retraction of the arm **164**, the angular orientation of the tines **106** is altered in a second rotational direction, opposite the first rotational direction, by pivoting of the tines **106** on the shafts **136**.

(72) FIGS. **23** through **25** show another example seed-double eliminator **200**. The seed-double eliminator **200** includes a rotatable wheel **202** disposed adjacent to the outer surface **57** of the seed disc **46**. In some implementations, the wheel **202** is passively operated in response to rotation of the seed disc **46**. In some implementations, the wheel **202** includes a plurality of protrusions **204** disposed radially about a center of the wheel **202**. In some implementations, the wheel **202** is formed from a rigid material, such as a rigid polymer (e.g., plastic), a metal, or a composite material. The wheel rotates on an arm **206** about an axis **208**. The arm **206** is pivotably mounted to a base **210** about an axis **212**. In some implementations, the axis **208** and the axis **212** are not parallel. This configuration allows the wheel **202** to ride along a contour of the outer surface **57** of the seed disc **46** as the seed disc **46** rotates. A spring **214** is provided between the base **210** and the arm **206** to bias the wheel **202** towards the seed disc **46** and away from the base **210**.

(73) As the seed disc **46** is rotated, recesses **216** formed in the outer surface **57** of the seed disc **46** engage with the protrusions **204** of the wheel **202** to cause the wheel **202** to rotate. Some of the

recesses **216** correspond to the raised features **68** and other correspond to the apertures **56**. As the wheel **202** rotates, the protrusions **204** extend into the apertures **56** and beyond the inner surface **55**. As a protrusion **204** extends into the aperture **56** (as shown in FIG. 25), seed retained at the aperture **56** is slightly disturbed. The protrusion **204** also affects a vacuum applied to the seed through the aperture **56**. As a result, of the disturbance caused by the protrusion **204**, any excess seed provided at an aperture **56** is dislodged and removed from the aperture **56**, leaving a single seed at the aperture **56**. Therefore, operation of the excess seed eliminator **800** operates to singulate seeds at each aperture **822**.

(74) In some implementations, the protrusions **204** extend beyond and is, thus, located inboard of the surface **55** by a distance **218**. In other implementations, the protrusion **204** is located outboard of the surface **55**. As illustrated in FIG. 45, in some instances, an end **215** of the protrusion **204** is positioned inboard of the surface **55** by a distance **218** when the protrusion **204** is aligned with the aperture **56**. In some instances, the distance **218** may be up to 1.0 mm (0.04 in.) when the protrusion **204** is aligned with the aperture **56**. In other instances, the end **215** of the protrusion **204** is located outboard of the surface by a distance **219**. In some instances, the distance **219** may be up to 0.5 mm (0.02 in.). In some instances, the end **215** of the protrusion **204** is flush with the surface **15** when the protrusion **204** is aligned with the aperture **56**. Thus, a position of the end **215** of the protrusion can be within a range 1.0 mm (0.04 in.) outboard of the surface **55** to 0.5 mm (0.02 in.) inboard of the surface **55**. As a result, an amount by which the protrusions **204** of the seed-double eliminator **200** extends beyond the inner surface **55** of the seed disc **46** is up to a factor of ten less than the amount by which the protrusions **88** of the ejector **76** extend beyond the inner surface **55** of the seed disc **46**.

(75) In some implementations, the protrusions **204** do not extend beyond the surface **55**. Rather, as explained earlier, the end **215** of the protrusions **204** may be flush with the surface **55** or be outboard of the surface **55** when the protrusions **204** are aligned with an aperture **56**. In such instances, the protrusions **204** disrupt the applied vacuum force to release excess seeds while reducing or eliminating an amount of direct contact between the protrusions **204** and the seeds located at the apertures **56**. Thus, in such instances, although the protrusions **204** do not extend inwardly beyond the surface **55**, the protrusions **204** may contact a portion of a seed extending outwardly beyond the surface **55**, disrupting a position of the seed relative to the aperture **56**. In this way, the protrusions **204** operate to unsettle or disrupt the seed doubles existing at an aperture **56** to release excess seeds while retaining a single seed at the aperture **56**. Additionally, the protrusions **204** occlude the open area of the apertures by between approximately 25% and 75%.

(76) Further, the seed double eliminator **200** is located along the seed disc **46** at a location between the seed pool **60** and the release point **62**. FIG. 46 is an oblique view of a portion of the seed meter **24**. As shown in FIG. 46, the seed double eliminator **200** is located between the seed pool **60** and the release point **62**, where the ejector **76** is located, along a seed path (i.e., a path followed by the seed from the seed pool **60** to the release point **62**). The seed double eliminator **200** works to singulate seeds at the apertures **56** as the seed disc **46** rotates and prior to the seeds reaching the ejector **76** where the seeds are separated from the seed disc **46** at the release position **62**.

Consequently, the seed double eliminator **200** is located upstream of the release point **62**. Thus, the seed double eliminator **200** differs from the ejector **76**, described above, in the function, the location along the seed disc **46**, and the amount of penetration of the respective protrusions beyond the inner surface **55** of the seed disc **46**. Particularly, the seed double eliminator **200** functions to unsettle seeds located at an aperture sufficiently to release excess seed (e.g., seeds in excess of a single seed at the apertures **56**) while retaining a single seed at the aperture as opposed to removing all seeds from an aperture **56** and depositing the seed into the seed transport apparatus **72** of the delivery system **26**, as performed by the ejector **76**. The seed-double eliminator **200** is located upstream of the delivery system **26** and is, therefore, not intended to displace seed onto the seed transport apparatus **72** for conducting seed to the ground. Additionally, the protrusions **88** of the

ejector **76** extend a larger distance beyond the inner surface **55** of the seed disc **46** to ensure separation of any seed located at an aperture **56** and, in some instances, obstruct a larger portion of the opening area of the apertures **56** to reduce an applied vacuum sufficiently to ensure separation of any seed at the apertures **56**. The protrusions **204** of the seed-double eliminator **200**, on the other hand, extend beyond the inner surface **55** of the seed disc **46** and obstruct a portion of the opening area of the apertures **56** merely to unsettle any seed present thereat to ensure separation of any excess seed and retain a single seed at the apertures **56**. Consequently, unlike the ejector **76**, the seed-double eliminator **200** operates to conserve seed, reduce costs, and improve the efficiency of a planting operation.

(77) FIGS. **47** and **48** show wheels **4700** and **4800** of seed double eliminators **4702** and **4802**, respectively. Each of the wheels **4700** and **4800** include protrusions that are different in shape. The wheel **4700** has protrusions **4704** similar to those of the wheel **202**. The protrusions **4704** have a flattened end **4706**. In the illustrated example, the protrusion **4702** occupies a majority of the opening area of the aperture **56**. As a result, the protrusion **4704** of the illustrated example reduces the vacuum force applied at the aperture **56**, which assists in separating excess seeds from the aperture **56**. In contrast, the protrusion **4804** of the wheel **4800** have a pointed or conical shape **4806**. The protrusion **4804** occupies a lesser amount of the opening area of the aperture **56**. Thus, the protrusions **4804** having this shape reduce the applied vacuum force at the apertures **56** to a lesser extent but still operate to unsettle and release excess seeds provided at the aperture **56**.

(78) FIGS. **26** and **27** illustrate another example seed-double eliminator **220**. The seed-double eliminator **220** is similar to the seed-double eliminator **200** except that the wheel **202** having the plurality of protrusions **204** is replaced with a deformable wheel **222**. In some implementations, the deformable wheel **222** lacks protrusions. In the illustrated the wheel **222** has a smooth surface. In other implementations, the wheel **222** has a textured surface. In some instances, the wheel **222** is formed from a deformable rubber or plastic material. In other instances, the wheel **222** can be formed from any material capable of conforming to a surface. Similar to the seed double eliminator **200**, the seed double eliminator **220** includes the wheel **222**, an arm **224**, and a base **226**. The wheel **222** is pivotably mounted to the arm **224**, and the arm **224** is pivotably mounted to the base **226**. A spring **228** is disposed between the arm **224** and the base **226** to apply a biasing force that urges the arm **224** and wheel **222** away from the base **226**. In the illustrated example, the spring **228** is a coil spring. In other implementations, the spring **228** can be other types of springs or components operable to urge the arm **224** to pivot away from the base **226**.

(79) A surface **230** of the wheel **222** is deformable, and, as the wheel **222** rotates along the outer surface **57** of the seed disc **46**, the surface **230** conforms to a topography or shape of the outer surface **57**, particularly in response to the biasing force provided by the spring **228**. In some implementations, as the surface **230** of the wheel **222** deforms, a portion of the wheel **222** extends into the recesses **216** formed in the outer surface **57** of the seed disc **46** and partially occludes or covers the apertures **56**, thereby reducing the applied vacuum force applied through the apertures **56**. As a result of the reduced vacuum force, excess seeds are released from the apertures **56** while retaining sufficient vacuum force to retain one of the seeds at the apertures **56**. Consequently, seed singulation is accomplished.

(80) FIGS. **28** through **34** illustrate a rotary seed double eliminator **300**. The seed double eliminator **300** includes a rotatable wheel **302** that is positioned adjacent both to the inner surface and the rim portion **54** of the seed disc **46**. In the illustrated example, the wheel **302** is rotatably coupled to a housing **304** disposed within the housing **30** via a shaft **305**. In some instances, the shaft **305** is a fastener (e.g., a bolt), a pin, or another component on which the wheel **302** can rotate. In some implementations, the shaft **305** rotates relative to the housing **304** via a bearing **307**.

(81) In some implementations, the housing **304** forms part of the housing **30**. In other implementations, the housing **304** is a separate component secured to or otherwise provided in the housing **30**. The wheel **302** is rotatable about an axis **306**. In some instances, the axis **306** is

perpendicular to an axis of rotation of the seed disc **46**, represented by line **308**.

(82) The wheel **302** includes a plurality of protrusions **310** having a bent shape. In some implementations, the protrusions **310** are radially extending from a peripheral edge of a hub portion **311** of the wheel **302**. A distal end portion **312** of the protrusions **310** is angled relative to a plane that is perpendicular to the axis **306**. As shown in FIG. **28**, the distal end portion **312** is angled towards the axis of rotation (line **308**) of the seed disc **46**. In some implementations, the distal end portion **312** encompasses an entirety of the protrusion **310**. In some implementations, the distal end portion **312** encompasses less than an entirety of the protrusion **310**. In some implementations, a surface **314** of the distal end portion **312** lies parallel to but offset from an adjacent portion of the inner surface **55** of the seed disc **46** when the protrusions **310** are located adjacent to the inner surface **55**, as shown in FIG. **28**.

(83) As the seed disc **46** rotates in the direction of rotation **58**, as shown in the series of images shown in FIGS. **30** through **34**, the protrusions **310** of the wheel **302** interact with the paddles **64** of the seed disc **46** to cause the wheel **302** to rotate in the direction of arrow **315**. Thus, the wheel **302** passively rotates in response to rotation of the seed disc **46**. This engagement causes the distal end portions **312** of the protrusions **310** to engage and disturb a seed or seeds retained on the seed disc **46** at the apertures **56**. In some implementations, in addition to physically disturbing the seed, the protrusions **310** sweep across a portion of the opening area **101** of the apertures **56**, obstructing a portion of the apertures **56** and causing a disruption to the applied vacuum force. This disruption unsettles the seed or seeds that are present at the apertures **56**. As a result, seeds in excess of a single seed are released and fall away from the aperture **56**, leaving a single seed retained at the aperture **56**.

(84) FIGS. **35** through **38** illustrate another example rotatable seed double eliminator **400**. The seed-double eliminator **400** includes a wheel **402** disposed adjacent to the inner surface **55** of the seed disc **46**. The wheel **402** is rotatable about an axis **404** and includes a plurality of radially extending protrusions **405**. The wheel **402** is located on a carriage **406** that is slidably coupled to a base **408**. The carriage **406** is movable along a surface **410** of the base **408** in the direction of arrows **412** and **414** in response to actuation of a position selector **416**. In some instances, the axis **404** is perpendicular to the surface **410**. In some implementations, the axis **404** forms an oblique angle relative to the inner surface **55**. Further, in some implementations, the axis **404** is disposed at an oblique angle relative to the axis of rotation of the seed disc **46**, represented by line **415**. The position selector **416** includes a dial portion **418** and a shaft **420**. The shaft **420** includes a threaded portion **422** and a groove **424**. In some implementations, the wheel **402** is rotatable on a shaft **407** extending from or through the carriage **406** and on a bearing **409** received onto an end **411** of the carriage **406**.

(85) In the illustrated example, the base **408** is fixed relative to the seed disc **46** and includes an end wall **418** a side wall **426** that define a cavity **428**. An opening **429** is formed at an end of the cavity **428** opposite the end wall **418**. The end wall **418** includes an opening **430** through which the shaft **420** of the position selector **416** extends. The threaded portion **422** includes threads **432** that engages mating threads **434** formed in the carriage **406**. In some implementation, the carriage **406** includes a carriage body **436** and an insert **438** disposed in a cavity **440** formed in the carriage body **436**. In some implementations, the carriage body **436** is formed from a polymeric material. In some implementations, the insert **438** is formed from a metal, and the threaded portion **422** of the position selector **416** is formed from a metal. In other implementations, one or both of the insert **438** and the threaded portion **422** is formed from a polymeric material. In some implementations, the insert **438** is omitted and the mating threads **434** are formed on an interior surface of the carriage body **432**. Other materials are also contemplated.

(86) The seed double eliminator **400** also includes a washer **442** disposed between a dial portion **418** of the position selector **416** and the end wall **418** and a retainer **444** disposed adjacent to the end wall **418** on a side opposite the washer **442**. The retainer **444** retains the position selector **416**

in position relative to the base **408**. In the illustrated example, the retainer **444** includes a threaded bore **446** that receives a set screw **448**. A portion of the set screw **448** is received within the groove **424** to maintain the position of the position selector **416** relative to the base **408**. The side wall **418** includes an opening **450** through which the set screw **448** is receivable into the retainer **444** and adjustable therein.

(87) In operation, as the position selector **416** is rotated in a first rotational direction, the threads **432** and mating threads **434** interact to cause the carriage **406** to move in the direction of arrow **412**. The carriage **406** is movable in the direction of arrow **414** in response to rotation of the position selector **416** in a second rotational direction, opposite the first rotational direction. A user can actuate the position selector **416** by rotating the position selector **416** via the dial portion **418**. Movement of the carriage **406** along the surface **410** in response to actuation of the position selector **416** includes moving the carriage **406** into and out of the cavity **438**. FIGS. **36** through **38** illustrate how an amount by which the protrusions **405** covers the opening area of the apertures **56** changes in response to operation of the position selector **416**. As shown in FIG. **36** through **38**, as the carriage **406** is moved in the direction of arrow **412**, an amount of the opening area of the apertures **56** swept by the protrusions **405** of the wheel **402** increases. As a result, an amount of engagement and disruption of any seeds located at the aperture **56** increases. As the carriage **406** is moved in the direction of arrow **414**, an amount of the opening area of the aperture **56** swept and obstructed by the protrusions **405** of the wheel **402** decreases, resulting in a reduced amount of engagement and disruption to any seeds located at the aperture **56**. Consequently, an amount of the opening area of the apertures **56** swept and obstructed by the protrusions **405** of the wheel **402** is adjustable, such as in response to a size or type of seed being transported by the seed disc **46**. Thus, an amount by which the protrusion **405** of the wheel **402** unsettles the one or more seeds located at the aperture **56** is adjustable using the position selector **416**. As a result of the disruption by the protrusions **405**, any excess seed at an aperture **56** is released, leaving a single seed at the apertures **56**. As shown in FIG. **35**, in some implementations, one or more of the apertures **56** has a tapered shape, with an opening size at the outer surface **57** larger than an opening size at the inner surface **56**.

(88) In other implementations, the position selector **416** is an actuator, such as a linear actuator or a rotary actuator. Other actuators are contemplated. In some instances, the actuator is an electrical actuator. In other instances, the actuator is powered by another type of power source, such as a hydraulic or pneumatic power source. For example, where the position selector **416** includes an electrical linear actuator, extension of the linear actuator (i.e., movement in a first linear direction) causes movement of the carriage **406** along the surface **410** in the direction of arrow **412**, and retraction of the linear actuator (i.e., movement in a second linear direction, opposite the first linear direction) causes movement of the carriage **406** along the surface **410** in the direction of arrow **414**.

(89) In some implementations, the wheel **402** also includes a conical or tapered surface **452**. The tapered surface **452** is disposed adjacent to the inner surface **55** of the seed disc **46**. In some instances, an angular displacement of the axis **404** from a perpendicular orientation relative to the inner surface **55** corresponds to an angle of the conical surface **452** relative to a plane perpendicular to the axis **404**. Orienting the wheel **402** relative to the inner surface **55**, as shown in FIG. **35**, provides for a portion of the tapered surface closest to the inner surface **55** remaining parallel to the inner surface **55** as the carriage **406** is moved along the surface **410** in response to actuation of the position selector **416** and as the wheel **402** rotates relative to the inner surface **55**.

(90) In some implementations, the wheel **402** is passively rotated in response to rotation of the seed disc **46**. The paddles **64** of the seed disc **46**, described earlier, engages with the protrusions **405** of the wheel **402** to rotate the wheel **402** as the seed disc **46** is rotated.

(91) FIGS. **39** and **42** illustrate another example seed double eliminator **500**. The seed double eliminator **500** includes a plurality of levers **502**. Although four levers **502** are illustrated, other implementations can include additional or fewer levers **502**, including a single lever **502**. The

levers **502** are located adjacent to the inner surface **55** of the seed disc **46** and are pivotable about a first end **504** of the lever **502**.

(92) The levers **502** are provided in recess **508** formed in a housing **510** located adjacent to the inner surface **55** of the seed disc **46**. In some implementations, the housing **510** extends circumferentially along a portion of the inner surface **55**. The levers **502** are pivotable, at the first end **504**, on a shaft **512** about respective axes **514**. In some implementations, the axes **514** are radially arranged relative to the axis of rotation of the seed disc **46**, represented by line **519**. A portion **516** of the first end **504** is received into a bore **518** formed in the housing **510**. The shaft **512** extends through at least a portion of the bore **518** and is received into a receptacle **520** formed in the first end **504** of the levers **502**. The levers **502** are pivotable between an extended position in which a second end **522** of the levers **502** extend beyond the recess **508** and a retracted position in which the second end **522** of the levers **502** are received into the recess **508**. The levers **502** are spring-loaded and biased in a first rotational direction **509** about the axes **514** towards the extended position. In the extended position, the second end **522** extend towards the apertures **56** formed in the seed disc **46**. In some implementations, the levers **502** are biased with a spring **507**, such as a coil spring or a torsion spring. However, other types of biasing components can be used to bias the lever **502**.

(93) The levers **502** are oriented such that an end surface **511** of the levers **56**, disposed adjacent to the inner surface **55**, are tangential to, parallel to, or are otherwise conforming to the inner surface **55** of the seed disc **46**. Further, in some implementations (for example, as shown in FIG. **39**), the axes **514** are perpendicular to the inner surface **55** of the seed disc **46**. The levers **502** also include a bearing surface **524**. The bearing surface **524** is engaged by an end **526** of the paddles **64**. When the paddles **64** begin to engage the bearing surface **524** at or proximate to the first end **504**, the levers **502** begin to pivot in a second rotational direction **525**, opposite the first rotational direction **509**, about the axes **514** towards the retracted position and away from the apertures **56**. As the paddles **64** continue to slide along the bearing surface **524**, the levers **502** continue to pivot about the axes **514** in the second rotational direction **525** towards the retracted position. Thus, an amount of pivoting of the levers **502** increases as the paddles **64** move an increased distance along the bearing surfaces **524**. The levers **502** reach the retracted position when the paddles **64** reach the second end **522** of the levers **502**. As the paddles **64** move past the second ends **522** of the levers **502**, a biasing force supplied by the biasing component rapidly moves the levers **502** back to the extended position, where the second ends **522** of the levers **502** contact and unsettle a seed or seeds located at the apertures **56** formed in the seed disc **46**. Further, in some implementations, when the second end **522** has returned to the extended position, the second end **522** obstructs an amount of the opening area **101** of the apertures **56**, thereby affecting the vacuum forced applied at the apertures **56**. As a result, excess seeds located at the apertures **56** are released, leaving a single seed resident at the apertures **56**.

(94) By including a plurality of levers **502**, if one lever **502** fails separate any excess seeds at an aperture **56**, engagement by the remaining levers **502** increases the likelihood that the excess seeds will be removed as the seed disc **46** continues to rotate. This benefit applies to other seed double eliminators described herein in which a plurality of the seed-engaging features are provided. However, as explained earlier, in some implementations, a single lever **502** may be included.

(95) FIGS. **41** and **42** show another example excess seed eliminator **600** that includes a plurality of spring strips **602**. Although three spring strips **602** are illustrated, other implementations can include additional or spring strips **602**, including a single spring strip **602**. The spring strips **602** form springs that are elastically deformable in response to engagement by the seed disc **46** and that spring back to an initial position after engagement between the seed disc **46** and the spring strips **602** ends.

(96) A first end **604** of the spring strips **602** is secured in a housing **606** positioned adjacent to the inner surface **55** of the seed disc **46** and extend in a cantilevered fashion. In some implementations,

the housing **606** extends circumferentially along a portion of the inner surface. Slots **608** are formed in the housing **606**, and the first end **604** of the spring strips **602** are received into a respective one of the slots **608** with a second end **610** provided on a freely extending portion **611**. The housing **606** also forms recesses **612** that extend from the slots **608**. The recesses **612** receive the freely extending portion **611** of the spring strips **602** when the spring strips **602** are deflected towards the housing **606** during operation of the seed disc **46**. In some implementations, the recesses **612** conform to a deflected shape of the spring strips **602**. The spring strips **602** are disposed adjacent to the inner surface **55** of the seed disc **46** and are biased into an extended position such that the second end **610** of the spring strips **602** is positioned adjacent to the apertures **56**. In some implementations, the spring strips **602** have an “S” shape and, in some instances, have a flattened “S” shape. However, in other implementations, the spring strips **602** have other shapes, such as a straight or planar shape.

(97) As the seed disc **46** rotates, the paddles **64** of the seed disc **46** engage and deflect the spring strips **602** towards the housing **606** such that the freely extending portion **611** of the spring strips **602** are received into the corresponding recesses **612** formed in the housing **606**. Particularly, an end **616** of the paddles **64** engages and slides along a bearing surface **618** of the spring strips **602**, causing elastic deformation of the spring strips **602** that deflects the spring strips **602** towards the housing **606** and causes freely extending portion **611** of the of the spring strips **602** to be received into the corresponding recesses **612**. Deflection of the spring strips **602** moves the freely extending portion **611** away from the apertures **56** and in a direction towards the outer edge **52** of the seed disc **46**. The deflection of the freely extending portion **611** can include rotation of at least a portion of the spring strip **602** about an axis **620**. In some implementations, the axis **620** is perpendicular to the inner surface **55** of the seed disc **46**. In some implementations, the axis **620** defines an oblique angle with the axis of rotation of the seed disc **46**, represented by line **622**. Further, an amount of deflection of the spring strips **602** increases as the paddles **64** move along the bearing surface **618** towards the second end **610**. A maximum amount of deflection occurs when the paddles **64** reach the second end **610** and just prior to the paddles **64** sliding off of the spring strips **602** at the second end **610**. When the paddles **64** move past the spring strips **602**, in response to an internal biasing force generated by the elastic deformation, the spring strips **602** spring back and return to the extended position. Extension of the spring strips **602** to the extended position causes the spring strips **602** (e.g., the second end **610** of the spring strips **602**) to contact and unsettle one or more seeds located at an aperture **56** formed in the seed disc **46**. In some instances, in the extended position, the spring strips **602** may partially obstruct the apertures **56**, affecting a vacuum force applied therethrough. For example, the freely extending portion **611** of the spring strips **602** may sweep across a portion of the opening area **111** of the apertures **56**, thereby obstructing the apertures **56**. As a result, an excess seed or seeds located at an aperture **56** are released, leaving a single seed located at the aperture **56**.

(98) FIGS. **43** and **44** are oblique views of another example excess seed eliminator **700**. The excess seed eliminator **700** includes a body **702** having a circumferential portion **704** and a conical portion **706** extending from the circumferential portion **704**. The conical portion **706** includes an edge **708** having a sawtooth shaped profile. The edge **708** includes plurality of sloped or slanted portions **710** defining a series of peaks **712** located at a trailing end **718** of the slanted portions **710**. Although a four slanted portions **710** are illustrated, in other implementations, additional or fewer slanted portion **710** may be used, including a single slanted portion **710**. The slanted portions **710** include a leading end **714** at a first end **716** and the trailing end **718** at a second end **720**, opposite the first end **716**. In implementations where the edge **708** includes a series of slanted portions **710**, the trailing end **718** of one slanted portion **710** is adjacent to and displaced from the leading end **714** of an adjacent slanted portion **710**.

(99) The excess seed eliminator **700** is stationary relative to a seed disc **722** and is mounted, for example, to a housing of a seed meter, which may be similar to the housing **30** of seed meter **24**.

The seed disc **722** is rotatable relative to the seed double eliminator **700**. The seed disc **722** is generally similar to seed disc **46** but may vary in some ways. For example, in some implementations, the seed disc **722** omits paddles similar to paddles **64**, paddles similar to paddles **104**, and raised features similar to raised features **68**. The seed disc **722** includes a tapered portion **723** that defines an inner surface **724** and an outer surface **726**. Raised portions **728** extend from the inner surface **724**. The raised portions **728** define surfaces **730** that are offset from the inner surface **724**, and an aperture **732** (similar to apertures **56**) extends through each raised portion **728**. The raised portions **728** hold seed **734** retained at the apertures **732** at an offset from the inner surface **724**, positioning the seed **734** to be engaged by the edge **708** and, particularly, the slanted portions **710** of the edge **708**. The conical portion **706** of the seed double eliminator **700** extends into an interior space **736** (similar to interior space **47**) of the seed disc **722** and is located adjacent to and offset from the inner surface **724**. Particularly, in some implementations, the conical portion **706** is or defines a surface that is parallel to the inner surface **724** of the seed disc **722**.

(100) The slanted portions **710** are angled relative to a plane disposed perpendicular to the axis of rotation **738** of the seed disc **722**. The slanted portions **710** are arranged such that, as the seed disc **722** rotates about the axis of rotation **738**, the leading end **714** (or a portion of the slanted portion **710** adjacent thereto) engages the seed **734** at the apertures **732**, and, as the seed disc **722** continues to rotate, an amount of disruption or dislocation of the seed **732** caused by the slanted portions **710** increases due to the angled nature of the slanted portions **710**. In some implementations, a maximum amount of disruption of the seeds **734** located at the apertures **732** occurs when the peaks **712** at the trailing end **718** of the slanted portions **710** reaches the apertures **732**. In some implementations, a slanted edge extends across more than one aperture simultaneously. For example, as shown in FIG. **44**, the slanted edge **710** has a length such that the slanted edge **710** extends across two apertures at the same time for a portion of a rotation of the seed disc **722**. This disruption upsets any excess seed in a seed double located at the aperture **732**. This disruption dislodges the excess seed, causing the excess seed to be released from the apertures **732**, leaving a single seed retained at the aperture **732**. The released seed, for example, falls back into the seed pool residing in the interior space **736** defined by the seed disc **722**. The seed double eliminator **700** having a series of slanted portions **710** provides for the ability to engage any excess seeds at a particular aperture **732** more than once to ensure separation of the excess seeds so as to provide a single seed at the aperture **732**.

(101) Without in any way limiting the scope, interpretation, or application of the claims appearing below, a technical effect of one or more of the example implementations disclosed herein is to conserve seed and reduce costs associated with a planting operation. Another technical effect of one or more of the example implementations disclosed herein is to improve the efficiency of a planting operation.

(102) Additional examples are also provided. A first example includes the following:

(103) A seed eliminator for an agricultural planter having a seed disc includes a base, an arm pivotably coupled to the base and pivotable about a first axis, and a rotatable wheel rotatably coupled to the arm. The arm is biased in a direction away from the base. The wheel is rotatable about a second axis and includes an exterior surface configured to unsettle a seed maintained at a location along a seed disc to remove the seed from the location.

(104) The seed double eliminator may also include a biasing component that biases the arm away from the base.

(105) The biasing component may be a spring.

(106) The exterior surface of the wheel may include a plurality of protrusions radially arranged on the wheel.

(107) The exterior surface may be deformable so as to conform, at least partially, to an abutting surface.

(108) An agricultural planter includes a seed disc rotatable about a first axis. The seed disc includes

a plurality of apertures formed along a perimeter thereof, and the apertures are configured to retain seed at the apertures adjacent to a first surface of the seed disc. The agricultural planter also includes a seed delivery system that includes a seed transport apparatus located adjacent to the first surface of the seed disc. The seed transport apparatus is movable along a continuous path to receive seed from the seed disc and transport the seed to an ejection point. The agricultural planter also includes a seed double eliminator disposed adjacent to a second surface opposite the first surface at a position that is upstream of a seed release location where seed retained at the apertures is released from the seed disc and captured by the seed transport apparatus of the transport assembly for conveyance to the ejection point. The seed double eliminator includes a base and an arm pivotably coupled to the base and pivotable about a second axis. The arm biased in a direction toward the seed disc. The seed double eliminator also includes a rotatable wheel rotatably coupled to the arm. The wheel is rotatable about a third axis and includes an exterior surface that engages the second surface of the seed disc and unsettles a seed retained at the apertures such that the seed is released from the seed disc.

(109) The wheel may be rotatable in response to rotation of the seed disc.

(110) The exterior surface of the wheel may include a plurality of radially extending protrusions. At least a portion of the protrusions may be received into the apertures to unsettle the seed retained at the apertures.

(111) Interaction between surfaces of the seed disc defining the apertures may impart a force to the plurality of protrusions to cause the wheel to rotate about the third axis.

(112) The surfaces defining the apertures may be conically shaped.

(113) A biasing component may bias the arm away from the base.

(114) The exterior surface of the wheel may include a plurality of protrusions radially arranged on the wheel.

(115) The exterior surface may be deformable so as to conform, at least partially, to the second surface of the seed disc.

(116) The exterior surface of the wheel may include a plurality of radially extending protrusions, and a portion of the protrusions may extend through the apertures and inwardly beyond the first surface of the seed disc.

(117) An end of at least one of the plurality of protrusions may extend beyond the first surface of the seed disc by an amount within a range of 0.25 mm (0.01 in.) to 0.5 mm (0.02 in.).

(118) The third axis may extend in a direction oblique to the first axis.

(119) A method of eliminating one seed of a seed double provided at a location along a seed disc includes disposing a wheel adjacent to an exterior surface of a rotatable seed disc at a location upstream of a seed release location where seed retained by the seed disc is released from the seed disc and captured by a seed transport apparatus of a transport assembly disposed adjacent to an inner surface of the seed disc for conveyance to an ejection point. The seed disc includes a plurality of apertures formed along a perimeter of the seed disc. The method also includes engaging the wheel with the exterior surface of the seed disc; rotating the wheel in response to a rotation of the seed disc; and unsettling a seed retained at one of the plurality of apertures of the seed disc by obstructing at least a portion of one of the apertures of the plurality of apertures with an exterior surface of the wheel.

(120) The method may also include biasing the wheel towards the seed disc.

(121) The exterior surface of the wheel may include a plurality of radially extending protrusions that are received, at least partially, into the plurality of apertures.

(122) Unsettling a seed retained at one of the plurality of apertures of the seed disc by obstructing at least a portion of one of the apertures of the plurality of apertures with an exterior surface of the wheel may include penetrating the one of the plurality of apertures with a protrusion of the wheel to cause the protrusion to extend beyond the inner surface of the seed disc by an amount within a range of 0.25 mm (0.01 in.) to 0.5 mm (0.02 in.).

(123) A second example includes the following:

(124) A seeding system for an agricultural planter that is operable to singulate seeds at one or more locations on a seed disc includes a seed disc rotatable about a first axis. The seed disc includes a rim portion, an inner surface, an outer surface opposite the inner surface, a plurality of apertures formed on the inner surface along the rim portion, and a plurality of paddles. Each aperture of the plurality of apertures defines an opening area, and the plurality of paddles is distributed among the plurality of apertures. The seeding system also includes a seed double eliminator disposed at a location along the rim portion of the seed disc and adjacent to the inner surface of the seed disc. The seed double eliminator is rotatable about a second axis in response to rotation of the seed disc. The seed double eliminator includes a wheel that includes a plurality of protrusions that sweep across at least a portion of the opening area of the plurality of apertures as the wheel rotates about the second axis.

(125) The plurality of protrusions may be radially extending protrusions.

(126) The wheel may include a hub portion defining a peripheral edge, and the plurality of protrusions may extend from the peripheral edge.

(127) At least one of the plurality of protrusions may have a bent shape.

(128) The bent shape may include a distal end portion that is angled inwardly towards the first axis.

(129) The distal tip may be parallel with a portion of the interior surface.

(130) The plurality of paddles may extend from the inner surface of the seed disc.

(131) The plurality of paddles may engage with the plurality of protrusions of the wheel to pivot the wheel about the second axis.

(132) The first axis may be perpendicular to the second axis.

(133) The plurality of protrusions may be configured to unsettle seed positioned at the plurality of protrusions as the plurality of protrusions sweep past the opening area as the wheel rotates about the second axis to cause a seed of a seed double to separate from the seed disc.

(134) A seed double eliminator for an agricultural planter having a seed disc includes a wheel rotatable about an axis. The wheel includes a plurality of protrusions extending radially from the axis, and each protrusion has a bent shape. The wheel is configured to be disposed adjacent to an interior surface of a seed disc and rotate in response to rotation of the seed disc and the protrusions configured to unsettle a seed retained at a location along the seed disc and release the unsettled seed from the seed disc.

(135) The bent shape of the protrusion may include a distal end portion that defines an angle that is oblique to the axis.

(136) A housing may be configured to be positioned adjacent to a rim portion of the seed disc, and the wheel may be configured to be positioned between the housing and the seed disc.

(137) The axis may be configured to extend perpendicularly to an axis of rotation of the seed disc.

(138) The wheel may be rotatable in response to a force applied to the protrusions.

(139) A method of eliminating excess seed carried by a seed disc of a seeding system includes disposing a seed double eliminator adjacent to a rim portion of a rotatable seed disc that is rotatable about a first axis. The seed double eliminator is rotatable about a second axis and includes a plurality of protrusions. The rotatable seed disc includes a plurality of apertures and a plurality of paddles. The plurality of apertures and the plurality of paddles have an alternating arrangement. The method also includes rotating the seed disc about the first axis; engaging the plurality of paddles of the seed disc with the plurality of protrusions of the seed double eliminator to rotate the seed double eliminator about the second axis; and sweeping the plurality of protrusions past the plurality of apertures to unsettle seed located at the plurality of apertures and release a seed of a seed double at at least one of the apertures.

(140) The plurality of protrusions may include a bent shape.

(141) The bent shape may include a distal tip portion that has defines an oblique angle relative to second axis such that the distal tip is parallel with an inner surface of the seed disc when the distal

tip portion is adjacent to the inner surface.

(142) Each of the plurality of apertures may include an opening area. Sweeping the plurality of protrusions past the plurality of apertures to unsettle seed located at the plurality of apertures and release a seed of a seed double at at least one of the apertures may include sweeping across, with the plurality of protrusions, at least a portion of an opening area as the seed double eliminator rotates about the second axis.

(143) The first axis and the second axis may be arranged perpendicular to each other.

(144) A third example includes the following:

(145) A seed double eliminator to release excess seed from a seed meter includes a base having a surface; a carriage movable along the surface; and a wheel coupled to the carriage and rotatable relative thereto about a first axis. The wheel includes a plurality of protrusions configured to unsettle and release a seed of a seed double from a seed meter. The seed double eliminator also includes a position selector operable engaged with the carriage to move the carriage along the surface of the base to selected position.

(146) The wheel may include a tapered surface.

(147) The tapered surface may be configured to be disposed adjacent to an inner surface of a seed meter such that a portion of the tapered surface disposed closest to the inner surface is parallel to the inner surface. The portion of the tapered surface may remain parallel with the inner surface when the carriage moves along the surface in response to operation of the position selector.

(148) The position selector may include an actuator, and actuation of the actuator in a first direction may cause the carriage to move in a second direction along the surface.

(149) Actuation of the actuator in a third direction may cause the carriage to move in a fourth direction along the surface, opposite the second direction.

(150) The axis may be perpendicular to the surface of the base.

(151) The position selector may include a first threaded surface. The carriage may include a second threaded surface that matingly engages the first threaded surface such that rotation of the position selector causes translational movement of the carriage along the surface.

(152) A seeding system for an agricultural planter operable to singulate seeds at one or more locations on a seed meter includes a seed meter rotatable about a first axis. The seed meter includes an inner surface, an outer surface opposite the inner surface, a plurality of apertures extending between the inner surface and the outer surface, and a seed double eliminator disposed adjacent to the inner surface of the seed meter. The seed double eliminator includes a base that includes a surface; a carriage movable along the surface; a position selector operably coupled to the carriage; and a wheel rotatably coupled to the carriage. The carriage is movable along the surface in response to operation of the position selector. The wheel is rotatable about a second axis. The wheel includes a plurality of protrusions, and the wheel is rotatable in response to rotation of the seed meter. The wheel is movable relative to the inner surface in response to movement of the carriage along the surface.

(153) The seed meter may include a peripheral portion, and the plurality of apertures may be formed in the peripheral portion.

(154) The wheel may include a tapered surface disposed adjacent to the inner surface, and a portion of the tapered surface that is disposed closest to the inner surface may be parallel with the inner surface.

(155) The second axis may be oblique to the inner surface.

(156) Each of the apertures of the plurality of apertures may define an opening area, and movement of the wheel relative to the inner surface in response to operation of the position selector may alter an amount of the opening area swept by the plurality of protrusions.

(157) Movement of the carriage in a first direction along the inner surface in response to operation of the position selector may cause the plurality of protrusions to sweep across an increased amount of the opening area of the apertures, and movement of the carriage in a second direction along the

inner surface, opposite the first direction, in response to operation of the position selector may cause the plurality of protrusions to sweep across a decreased amount of the opening area of the apertures.

(158) The plurality of protrusions may sweep across at least a portion of the opening area of the apertures when the wheel rotates about the second axis to unsettle and separate a seed of a seed double located at an aperture of the plurality of apertures.

(159) The position selector may be an actuator. A first mode of operation of the actuator may cause the carriage to move along the surface in a first direction, and a second mode of operation of the actuator may cause the carriage to move along the surface in a second direction, opposite the first direction.

(160) The seed meter may include a plurality of paddles formed on the inner surface, and the plurality of paddles may engage the plurality of protrusions to cause the wheel to rotate about the second axis in response to rotation of the seed meter about the first axis.

(161) The plurality of paddles and the plurality of apertures may have an alternating arrangement.

(162) The first axis may be disposed at an oblique angle relative to first axis.

(163) A method of eliminating excess seed carried by a seed meter of a seeding system includes providing a seed double eliminator adjacent to an inner surface of a seed meter rotatable about a first axis. The seed double eliminator includes a base comprising a surface, a carriage movable along the surface, and a wheel that includes a plurality of protrusions. The wheel is coupled to and rotatable relative to the carriage about a second axis in response to rotation of the seed meter. The method also includes rotating the seed meter about the first axis; engaging the plurality of protrusions of the wheel with a plurality of paddles of the seed meter to cause rotation of the wheel; sweeping the plurality of protrusions past a plurality of apertures formed in the seed meter in response to rotation of the wheel; and engaging seed at the plurality of apertures with the plurality of protrusions to unsettle seed located at the plurality of apertures and release a seed of a seed double at at least one of the apertures.

(164) The method may also include altering a position of the carriage relative to the surface of the base to alter an amount by which the plurality of protrusions sweep across the plurality of apertures.

(165) A fourth example includes the following:

(166) A seeding system includes a seed disc rotatable about a first axis. The seed disc includes an inner surface, an outer surface opposite the inner surface, and a plurality of apertures extending between the inner surface and the outer surface. Each aperture defines an opening area. The seeding system also includes a seed double eliminator that includes a movable lever disposed adjacent to the inner surface of the seed disc. The moveable lever includes a first end and a second end. The moveable lever is pivotable at the first end about a second axis and biased in a first rotational direction about the second axis towards the plurality of apertures. The second end of the moveable lever is movable past at least a portion of the opening area of the apertures.

(167) The seed disc may include a plurality of paddles that engage the movable lever. The plurality of paddles may engage the movable lever to cause the movable lever to pivot in a second rotational direction about the second axis, away from the plurality of apertures, as a paddle of the plurality of paddle moves across the movable lever. The movable lever may pivot about the second axis in the first rotational direction, opposite the first rotational direction, towards the plurality of apertures when the paddle of the plurality of paddles ceases engagement with the movable lever.

(168) The plurality of apertures and the plurality of paddles may have an alternating arrangement.

(169) The moveable lever may include a bearing surface extending from the first end to the second end, and the plurality of paddles may slide along the bearing surface to cause the moveable lever to pivot in the second rotational direction.

(170) An amount of rotation of the moveable lever about the second axis in the second rotational direction may increase as the plurality of paddles moves across the bearing surface of the movable

lever.

(171) A housing may be disposed adjacent a peripheral edge of the seed disc, and the moveable lever may be pivotably attached to the housing.

(172) The seed double eliminator may include a plurality of movable levers and a plurality of second axes. Each moveable lever of the plurality of moveable levers may be pivotable about a respective one of the second axes, and the second axes may be radially arranged relative to the first axis.

(173) The second axis may be perpendicularly oriented relative to the inner surface.

(174) The seed double eliminator may include a spring, and the spring may bias the moveable lever in the first direction.

(175) A method of eliminating excess seed carried by a seed disc of a seeding system includes providing a seed double eliminator adjacent to an inner surface of a seed disc rotatable about a first axis. The seed double eliminator includes a lever having a first end and a second end. The lever is pivotable at the first end about a second axis. The method also includes rotating the seed disc about the first axis; engaging the lever with a paddle of the seed disc to pivot the lever away from an initial position in a first direction; and engaging seed at an aperture formed within the seed disc with the lever to release a seed of a seed double at the aperture.

(176) The method may include pivoting the lever in a second direction, opposite the first direction, when the paddles ceases to engage the lever.

(177) Pivoting the lever in a second direction, opposite the first direction, when the paddles ceases to engage the lever may include returning the lever to the initial position.

(178) The method may also include biasing the lever in the second direction toward the aperture.

(179) Biasing the lever in the second direction toward the aperture may include biasing the lever with a spring.

(180) The method may also include sweeping the lever across at least a portion of the opening area of the aperture formed in the seed disc.

(181) The second axis may be perpendicular to the inner surface of the seed disc.

(182) The second axis may extend obliquely relative to the first axis.

(183) Engaging seed at the aperture formed within the seed disc with the lever to release the seed of a seed double at the aperture may include rapidly pivoting the lever in the second direction in response to cessation of engagement with the seed disc to impact at least one of the seed double.

(184) A seed double eliminator to release excess seed from a seed disc includes a lever. The seed double eliminator includes a first end, a second end, and a surface extending between the first end and the second end. The lever is pivotable about an axis at the first end in a first direction and a second direction, opposite the first direction. The lever is biased in the first direction at a first position and configured to be displaced in the second direction in response to engagement with a seed disc and to impact a seed double when the lever is returned the first position to release a seed of the seed double.

(185) The lever further may include a surface configured to be engaged by a portion of the seed disc to cause the lever to rotate an increasing amount in the second direction as the portion of the seed disc slides along the surface.

(186) A fifth example includes the following:

(187) A seeding system includes a seed disc rotatable about a first axis. The seed disc includes an inner surface, an outer surface opposite the inner surface, and a plurality of apertures extending between the inner surface and the outer surface. Each aperture defines an opening area. The seeding system also includes a seed double eliminator that includes a resilient spring strip fixed at a first end. The spring strip is disposed adjacent to the inner surface of the seed disc, and the spring strip is elastically deformable in response to deflection by the seed disc.

(188) The spring strip may be S-shaped.

(189) The seed disc may also include a plurality of paddles formed on the inner surface. Contact

between plurality of paddles and the spring strip may cause deflection of the spring strip.

(190) The seed disc may include an outer edge, and deflection of the spring strip in response to contact between the plurality of paddles and the spring strip may include deflection of the spring strip towards the outer edge.

(191) The plurality of paddles and the plurality of apertures may have an alternating arrangement.

(192) The spring strip may include a bearing surface, and the plurality of paddles may engage the bearing surface. An amount of deflection of the spring strip in response to engagement between the plurality of paddles and the bearing surface may increase as the plurality of paddles move across the surface as the seed disc rotates.

(193) The spring strip may include a bearing surface. The plurality of paddles may engage the bearing surface. The spring strip may be located at a first position when not engaged with any paddle of the plurality of paddles, and the spring strip may spring back to the initial position when a paddle of the plurality of paddles slides off of the bearing surface.

(194) The elastic deformation of the spring strip in response to deflection by the seed disc may include rotation of a of at least a portion of the spring strip about a second axis.

(195) The second axis may be perpendicular to the inner surface of the seed disc.

(196) The first axis and the second axis may define an oblique angle.

(197) The seed double eliminator may include a housing disposed adjacent to the inner surface of the seed disc. The housing may include a slot, and the first end of the spring strip may be received into the slot.

(198) The housing may include a recess extending from the slot, and the recess may conform to a shape of the spring strip when the spring strip is deflected by the seed disc.

(199) The recess may include a curved surface.

(200) Each of the plurality of apertures may include an opening area, and the spring strip may sweep across at least a portion of the opening area as the seed disc rotates.

(201) A method of eliminating excess seed carried by a seed disc of a seeding system includes providing a seed double eliminator adjacent to an inner surface of a seed disc rotatable about a first axis. The seed double eliminator includes a cantilevered strip that includes a fixed end and a freely extending end. The method also includes rotating the seed disc about the first axis; engaging the cantilevered strip with a paddle of the seed disc to elastically deform the cantilevered strip; and engaging seed at an aperture formed within the seed disc with the cantilevered strip to release a seed of a seed double at the aperture.

(202) Engaging seed at an aperture formed within the seed disc with the cantilevered strip to release a seed of a seed double at the aperture may include springing back the cantilevered strip when the paddles moves past the cantilevered strip and impacting the seed to release the seed of the seed double at the aperture.

(203) Engaging the cantilevered strip with a paddle of the seed disc to elastically deform the cantilevered strip may include contacting a surface of the cantilevered strip extending from the fixed end to the freely extending end with the paddle.

(204) The method may also include elastically deflecting the cantilevered strip with the paddle of seed disc and receiving the deflected cantilevered strip into a recess formed within a housing.

(205) The cantilevered strip may be S-shaped.

(206) The aperture may include an opening area, and the method may also include sweeping across a portion of the opening area with a portion of the cantilevered strip.

(207) A sixth example includes the following:

(208) A seeding system includes a seed disc rotatable about an axis. The seed disc includes an inner surface, an outer surface, a peripheral edge, and a plurality of apertures radially arranged and extending between the inner surface and the outer surface. Each aperture of the plurality of apertures includes an opening area. The seeding system also includes a seed double eliminator disposed adjacent to the inner surface. The seed double eliminator includes a portion that is parallel

to an adjacent portion of the seed disc. The portion of the seed double eliminator defines a slanted edge that is oblique relative to the peripheral edge. The portion of the seed double eliminator is positioned such that the portion of the seed double eliminator covers an increasing amount of the opening area of the aperture as the seed disc rotates relative to the seed double eliminator about the axis.

(209) The portion of the seed double eliminator may include a sawtooth shaped profile that includes a plurality of the slanted edges. The plurality of slanted edges may be arranged in series along the inner surface of the seed disc.

(210) A peak may be arranged at an end of each of the plurality of slanted edges.

(211) The portion of the seed double eliminator may extend along at least a portion of the inner surface of the seed disc.

(212) The slanted edge may extend across a more than one aperture of the plurality of apertures.

(213) The seed disc may include a conical portion, and the portion of the seed double eliminator may conform to but be offset from the conical portion of the seed disc.

(214) The seed disc further may include a plurality of raised portions radially arranged along the inner surface, and an aperture of the plurality of apertures may extend through at least one of the plurality of raised portions.

(215) Each raised portion may include a raised surface, and the portion of the seed double eliminator may conform to the raised surface when the portion of the seed double eliminator is located adjacent to the raised surface.

(216) A method of eliminating excess seed carried by a seed disc of a seeding system includes locating a seed double eliminator adjacent to an inner surface of a seed disc that includes a plurality of apertures and is rotatable about an axis of rotation. The seed double eliminator includes an edge that has a sloped portion that defines an oblique angle relative to a plane perpendicular to the axis of rotation of the seed disc. The method also includes rotating the seed disc relative to the seed double eliminator and engaging seed at an aperture formed within the seed disc with the sloped portion of the edge to release a seed of a seed double at the aperture.

(217) Rotating the seed disc relative to the seed double eliminator may include rotating the seed disc about the axis of rotation.

(218) The edge may include a plurality of the sloped edges, and the plurality of sloped edges may be arranged in series along the inner surface of the seed disc.

(219) A peak may be arranged at an end of each of the plurality of sloped edges.

(220) The portion of the seed double eliminator may extend along at least a portion of the inner surface of the seed disc.

(221) The seed double eliminator may include a conical portion, and the edge is formed on the conical portion.

(222) The seed disc may include a plurality of raised portions radially arranged along the inner surface, and an aperture of the plurality of apertures may extend through at least one of the plurality of raised portions.

(223) Each raised portion may include an interior surface, and the conical portion of the seed double eliminator may conform to the interior surface when the portion of the seed double eliminator is located adjacent to the raised portions.

(224) Each of the apertures may define an opening area, and rotating the seed disc relative to the seed double eliminator may include causing the plurality of apertures to pass the edge of the seed double eliminator and progressively cover an increasing amount of the opening area of the apertures with the sloped portion of the edge.

(225) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed, but rather as descriptions of features that may be specific to particular implementations. Certain features that are described in this specification in the context of separate implementations can also be implemented, in combination,

in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations, separately, or in any suitable sub-combination. Moreover, although previously described features may be described as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can, in some cases, be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

(226) Particular implementations of the subject matter have been described. Other implementations, alterations, and permutations of the described implementations are within the scope of the following claims as will be apparent to those skilled in the art. While operations are depicted in the drawings or claims in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed (some operations may be considered optional), to achieve desirable results. In certain circumstances, multitasking or parallel processing (or a combination of multitasking and parallel processing) may be advantageous and performed as deemed appropriate.

(227) Accordingly, the previously described example implementations do not define or constrain the present disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of the present disclosure.

Claims

1. A seed double eliminator for a seeding machine having a seed disc, the seed double eliminator comprising: a housing comprising: a cavity; and a shaft formed in the cavity, the shaft defining a pivot axis; a tine mounted on the shaft and pivotable about the pivot axis; and a selector operably engaged with the tine, the selector movable to alter an angular orientation of the tine about the pivot axis.
2. The seed double eliminator of claim 1, wherein the selector is movable to cause movement of the tine about the pivot axis in discrete amounts.
3. The seed double eliminator of claim 1, wherein the selector is movable to cause movement of the tine over a predetermined range.
4. The seed double eliminator of claim 3, wherein the range is approximately 30° of rotation.
5. The seed double eliminator of claim 1, further comprising a frame extending between the tine and the selector, the frame movable in response to movement of the selector to cause movement of the tine about the pivot axis.
6. The seed double eliminator of claim 5, wherein the tine comprises: an elongated portion; a body portion; and a protrusion extending from the body portion, wherein the body portion defines an aperture that receives the shaft, wherein the frame comprises an opening, and wherein the protrusion is received into the opening of the frame.
7. The seed double eliminator of claim 6, wherein the tine is movable about the pivot axis in response to a moment imparted to the protrusion by the frame.
8. The seed double eliminator of claim 7, wherein the opening defines an edge of the frame, and wherein the edge of the frame imparts the moment to the protrusion to alter the angular orientation of the tine about the pivot axis.
9. The seed double eliminator of claim 5, wherein the selector is rotatable and wherein rotation of the selector causes movement of the frame along a path.
10. The seed double eliminator of claim 9, wherein the selector comprises a first gear, wherein the frame comprises a second gear intermeshed with the first gear, and wherein rotation of the selector causes the first gear to rotate the second gear which, in turn, causes movement of the frame along the path.
11. The seed double eliminator of claim 5, wherein the selector comprises a powered actuator, wherein actuation of the actuator causes movement of the frame along a path.

12. The seed double eliminator of claim 5, wherein the frame is disposed in a channel, and wherein movement of the selector causes the frame to move along a curved path within the channel.
 13. An agricultural planter comprising: a seeding machine configured to singulate seeds prior to depositing the seeds into the ground, the seeding machine comprising: a rotatable seed disc; and a seed double eliminator disposed adjacent to an inner surface of the seed disc, the seed double eliminator comprising: a housing disposed along a perimeter of the seed meter, the housing comprising: a cavity; and a shaft extending into the cavity and defining a pivot axis; a tine mounted on the shaft and pivotable about the pivot axis; and a selector operably engaged with the tine, the selector movable to alter an angular orientation of the tine about the pivot axis.
 14. The agricultural planter of claim 13, wherein the seed double eliminator further comprises a frame extending between the tine and the selector, the frame movable in response to movement of the selector to cause movement of the tine about the pivot axis.
 15. The agricultural planter of claim 14, wherein the frame is disposed in a channel, and wherein movement of the selector causes the frame to move along a curved path within the channel.
 16. The agricultural planter of claim 14, wherein the tine comprises: an elongated portion; and a body portion that includes a protrusion and defines an aperture that receives the shaft, wherein the frame comprises an opening, wherein the protrusion is received into the opening of the frame, and wherein the tine is movable about the pivot axis in response to a moment imparted to the protrusion by the frame.
 17. The agricultural planter of claim 14, wherein the selector is rotatable and wherein rotation of the selector causes movement of the frame along a path.
 18. The agricultural planter of claim 14, wherein the selector comprises a powered actuator and wherein actuation of the actuator causes movement of the frame along a path.
 19. A method for altering a rotational orientation of a seed double eliminator, the method comprising: operating a selector of a seed double eliminator, the seed double eliminator comprising: the selector; a frame operably engaged with the selector; and a tine pivotably movable about a first axis of rotation, the frame operably engaged with the tine to cause the pivotable movement of the tine; moving the frame in response to operation of the selector; and imparting a moment to the tine by the frame to pivot the tine about the first axis of rotation.
 20. The method of claim 19, wherein operating a selector comprises rotating the selector about a second axis of rotation.
 21. The method of claim 19, wherein moving the frame in response to operation of the selector comprises moving the frame along a path in response to operation of the selector.
 22. The method of claim 19, wherein the selector comprises a powered actuator operably engaged with the frame, wherein operating the selector comprises actuating the actuator, and wherein moving the frame in response to operation of the selector comprises displacing the frame in response to actuation of the actuator.
 23. The method of claim 19, wherein the frame comprises an aperture, wherein a protrusion of the tine is received into the aperture, and wherein imparting a moment to the tine by the frame to pivot the tine about the second axis of rotation comprises imparting a force to the protrusion by surface of the frame defining the aperture.
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